Valencia Balam Susanto

+852 5972 8389 | valenciabalamm@gmail.com | valenciabalamsusanto.github.io

EDUCATION

The Hong Kong Polytechnic University
BEng (Hons) in Electronic Systems & Internet-of-Things *2023 — 2027*

- Expected **First-Class Honours** | FEng Dean's List | Recipient of PolyU Post Entry Scholarship
- Related Subjects: AI & Data Analytics (A+), Electronic System Training (A-), Machine Learning (A+)

WORK EXPERIENCE

MTR Corporation Ltd Hong Kong
Information Technology & AI Intern *Jul — Aug 2025*

- Led end-to-end development of **2 internal RAG chatbots**, from requirements, prompt engineering, to deployment. Increased both enterprise AI adoption and workplace **efficiency by 20%+**.
- Built LLM-as-Judge **evaluation tool via Python** that acted as QA automation for RAG chatbots, cutting turnaround time and errors by **50%+** and manual testing effort by **30%+.**
- Presented product demos to technical and business stakeholders, including the CIO.

BeeInventor Ltd Hong Kong
Information Engineering Intern *Jun — Aug 2024*

- Analysed large-scale temperature data using **Python, SQL, and Excel**. Successfully identified sensor calibration issues and ~5 helmets with faulty temperature sensor in active use.
- Refined sensor software and embedded C firmware, improved temperature reading **accuracy by 5%+.**
- Researched IoT industry developments to support R&D decisions.

The Hong Kong Polytechnic University Hong Kong
Part-time Research Student Assistant *Sep 2023 — Present*

- Support Dr. Jeffry Oktavianus (Assistant Professor) **across 5+ research projects** in data collection, dataset management and exploratory AI research.
- Collect and clean research data in Excel and Python, **delivering 3+ structured, analysis-ready datasets** with summaries of key insights and trends.

PROJECTS

Large Language Models | *Application in Bahasa Indonesia Scam Detection (Final Year Project)*

- Building an automated Python experiment pipeline that batch-processes 200+ scam call transcripts, replacing manual prompting to compare 3 LLMs and RAG setups quickly and repeatably.

Machine Learning | *Automated Loan Approvals & Default Prediction*

- Built approval and default-risk models in Python on 1.6M Lending Club applications and 546K loans, using XGBoost, scorecards and SHAP. Trained a machine learning model that out-ranked Lending Club's own grades on later loans (**Gini 0.385 vs 0.357**).

Exploratory Data Analysis | *Sales & Market Analysis of Electronic Products on Amazon*

- Cleaned 42,675 Amazon listings into 8,149 electronics products for pricing, discounts, trends and market basket analysis. **Built an interactive dashboard** to consolidate findings and guide seller decisions.

SKILLS

- **Languages:** English, Indonesian, Chinese
- **Programming:** Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, Matplotlib, Seaborn), SQL, C, C++, HTML, CSS, Javascript
- **Data & AI:** Excel, Power BI, Tableau, Predictive Modelling, RAG, LLM Applications & Evaluation

VOLUNTEERING

- English teacher at IMWU, supporting Indonesian migrant domestic workers in Hong Kong.
- Taught STEM and AI to underprivileged children on two separate outreach programmes, in Indonesia and the Philippines respectively.